

## Practical Thinking: Data Centre Power Supply Section – Equipped with Diotec SiC MOSFET DI065SIC031TL



In modern data centers, power supply efficiency and reliability are critical for ensuring continuous operation of servers, storage systems and networking equipment. Data center power supplies must handle high power loads while minimizing energy loss and heat generation, as even small inefficiencies can lead to significant operational costs. SiC MOSFETs play a key role in high-frequency switching power supplies, enabling efficient AC-DC conversion and voltage regulation for stable power delivery. Their fast switching capability, low conduction losses and high thermal performance are essential for maintaining system reliability in demanding 24/7 operating environments

In data center power supply applications, SiC MOSFETs act as the primary switching elements in power conversion stages such as Power Factor Correction (PFC) and DC-DC converters. Low  $R_{DS(on)}$ , fast switching speed and excellent thermal stability are crucial parameters for achieving high efficiency and reducing power dissipation. These characteristics help improve overall power density, reduce cooling requirements and ensure long-term performance, making MOSFETs an essential component for next-generation energy-efficient data center infrastructure.

To meet these condition, the DI065SIC031TL from Diotec Semiconductor is an 128 A/650 V N-channel SiC MOSFET ready to deliver highest Voltage capability is up to 650V, requiring less Series connection. The low on-state resistance of typically 31 mOhm helps improve the heat dissipation performance of SiC MOSFET, reduce the volume and improve the power density of SiC MOSFET.

FOR MORE INFO



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